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Quality Diversity under Sparse Interaction and Sparse Reward: Application to Grasping in Robotics

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Abstract

Quality-Diversity (QD) methods are algorithms that aim to generate a set of diverse and high-performing solutions to a given problem. Originally developed for evolutionary robotics, most QD studies are conducted on a limited set of domains – mainly applied to locomotion, where the fitness and the behavior signal are dense. Grasping is a crucial task for manipulation in robotics. Despite the efforts of many research communities, this task is yet to be solved. Grasping cumulates unprecedented challenges in QD literature: it suffers from reward sparsity, behavioral sparsity, and behavior space misalignment. The present work studies how QD can address grasping. Experiments have been conducted on 15 different methods on 10 grasping domains, corresponding to 2 different robot-gripper setups and 5 standard objects. An evaluation framework that distinguishes the evaluation of an algorithm from its internal components has also been proposed for a fair comparison. The obtained results show that MAP-Elites variants that select successful solutions in priority outperform all the compared methods on the studied metrics by a large margin. We also found experimental evidence that sparse interaction can lead to deceptive novelty. To our knowledge, the ability to efficiently produce examples of grasping trajectories demonstrated in this work has no precedent in the literature.

Keywords

Quality diversity, Sparse reward, Sparse behavior, Grasping, Evolutionary robotics.

1 Introduction

Quality-Diversity (QD) methods are evolutionary algorithms that optimize both diversity and quality to generate large repertoires of high-performing solutions to a given problem (Pugh et al. (2016), Cully and Demiris (2017)). This field produced significant results in evolutionary robotics, including recovery from injury (Cully et al. (2015)), generation of adversarial objects for robotic grasping (Morrison et al. (2020)), or morphological evolution (Zardini et al. (2021)). The recent rise of interest led to novel interactions between fields, with notable combinations

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- We propose a taxonomy to avoid ambiguities when talking about QD methods, especially regarding NS-related methods;
- We discuss the role of the behavioral characterization through the notions of *behavioral alignment*, *density of the behavioral function*, *behavioral completeness* and *driving/describing behavior space*;
- We introduce a simple framework that distinguishes the evaluation of a QD method from its internal components;
- We show that a simple variant of MAP-Elites consistently dominates state-of-the-art QD methods on the considered metrics, demonstrating capabilities to generate a large set of diverse and high-performing solutions on grasping – despite this task challenges: sparse fitness, sparse interaction, and behavior misalignment;
- We investigate the impact of the behavioral sparsity on QD methods performances, obtaining empirical evidence that sparse interaction can lead to deceptive novelty.

The code is available on Github¹. We believe these results will open the way to apply QD on more complex tasks related to robotic manipulation, eventually solving problems that cannot easily be tackled with learning methods from other fields. The experimental results demonstrated here show that **QD methods can efficiently be leveraged to generate grasping trajectories of different fitnesses**. Such data could be used to bootstrap learning strategies of any kind. Generating grasping demonstrations is a **key matter to solve this task** (Wang et al. (2021), De Coninck et al. (2020)). To our knowledge, **no method in the literature is able to easily produce examples of grasping trajectories on different robots and objects as shown in this work**.

2 Related works

2.1 Quality diversity

While standard optimization approaches search for the extremum solution to a single-objective solution, Quality-Diversity (QD) algorithms aim to generate a set of diverse and high-performing solutions. Those methods lead to application in many fields, including image generation (Fontaine and Nikolaidis (2023)), discovery of drugs (Verhellen and Van den Abeele (2020)), or engineering optimization (Gaier et al. (2018)). QD methods rely on a behavioral characterization to compare the evaluated solutions for a given task, allowing to maintain diversity along with the optimization of a quality criterion.

QD methods emerged through two seminal works, NSLC and MAP-Elites. NSLC (Lehman and Stanley (2011b)) is a population-based evolutionary method that adds pressure toward the most novel and high-performing individuals through a Pareto-front selection. It has first been introduced as an extension of Novelty Search (NS) (Lehman and Stanley (2011a)) – an approach that replaces the quality-guided optimization process with a novelty-guided one. MAP-Elites (ME) (Mouret and Clune (2015)) is the second seminal QD method. It relies on a structured container that keeps the best previously generated solutions for different behavioral niches. Almost all QD algorithms derive from those two pioneer methods. However, MAP-Elites-based algorithms seem to be the most popular ones: most of the current state-of-the-art methods for rapid illumination of a behavior space are more or less complex variants of ME (Fontaine and Nikolaidis (2023), Macé et al. (2023), Faldor et al. (2023)).

¹https://github.com/Johann-Huber/qd_grasp

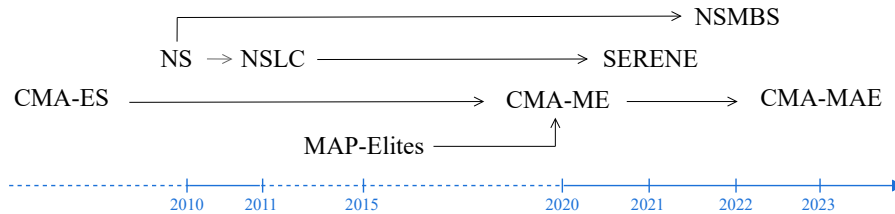


Figure 1: Timeline of compared methods. The literature is split into NS-derived methods and MAP-Elites-derived methods. In this work, we compare the 3 most promising methods for rapid illumination of a behavior space in sparse reward and behavior (SERENE, NSMBS, and CMA-MAE) with 12 other methods mainly derived from NS and ME. To fairly compare them, we proposed an evaluation framework in which the collected metrics are not dependent on the algorithms’ components.

Most of the QD works in robotics are actually applied to locomotion (Lehman and Stanley (2011b), Macé et al. (2023), Faldor et al. (2023), Zardini et al. (2021)). In these domains, it is easy to design a behavioral characterization that expresses the agent’s displacement, such that the exploration of this behavior space will eventually lead to the optimal solution (Lehman and Stanley (2011a), Paolo et al. (2021)). The present work aims to show that QD can efficiently be applied to more complex tasks like grasping, despite the involved challenges: sparse fitness, sparse interaction, and misaligned behavior space (see section 3.3.1).

The present work calls *complete* a behavioral characterization that allows an easy exploration by guaranteeing that a successful solution will eventually be found through the use of a non-constant behavioral signal (see section 3.3.1.4). This paper shows that **QD methods can efficiently be applied on more complex tasks like robot manipulation, in which the algorithm cannot rely on a complete behavioral characterization.**

2.2 QD for hard exploration problems

2.2.1 NS and ME divergence

Novelty Search has been introduced as a promising approach to address sparse or deceptive reward problems (Lehman and Stanley (2011a)). This seminal paper led to the emergence of Quality Diversity, as NSLC, the first QD method, derives from NS. Interestingly, the QD field has led to two research branches: **NS-based** and **ME-based works** (Figure 1). NS and ME-based methods are rarely compared in works involving one or the other family (Fontaine and Nikolaidis (2023), Kim et al. (2021)). All those methods involve similar properties and mechanisms; we argue that **comparing them might lead QD practitioners to insightful results.**

2.2.2 QD for sparse reward domains

Recent works have studied how QD methods could address domains submitted to sparse rewards: SERENE (Paolo et al. (2021)) proposes a new approach to optimize a real-valued fitness function in the sparse context; NSMBS (Morel et al. (2022)) explores multiple behavior spaces to generate grasping trajectories; CMA-MAE (Fontaine and Nikolaidis (2023)) claims to fix the limitations of CMA-ME (Fontaine et al. (2020)) on flat fitness landscapes, and reported state-of-the-art results on challenging tasks. Those methods can be considered as the most promising algorithms for addressing the task of grasping, which is submitted to sparse reward: SERENE is the only QD method that explicitly does Rapid Illumination of a Behavior Space (RIBS) in

sparse reward, NSMBS is the only QD method that demonstrated results on grasping, and CMA-MAE is a state-of-the-art method for doing RIBS, getting specific algorithmic mechanism to be more robust to sparse reward domains. **The present works compared these 3 methods with 12 other algorithms to identify the best-performing approach for doing RIBS on grasping.**

2.3 Grasping

Grasping refers to making an agent solidarize its end effector with an object by applying forces and torques on its surface. This task is of great interest to robotics and artificial intelligence research communities, as it is considered a prerequisite for many manipulation tasks (Hodson (2018)). After the early ages of analytical-based methods (Nguyen (1988)), data-driven approaches have dominated the literature on grasping in robotics since the beginning of the 21st century (Zhang et al. (2022)). Despite the involved research efforts, **grasping is still partially solved**: the reward sparsity of grasping makes it very challenging for learning methods to generate data to bootstrap learning from. To increase the chances of success of random movements, most of the approaches constrain the operational space to top-down movements (Yang et al. (2023)) or are limited to parallel grippers (Fang et al. (2020)). As the self-supervised acquisition of data is very expensive (Levine et al. (2018)), most of the recent works on grasping rely on human-provided demonstrations (Wang et al. (2021), Mosbach and Behnke (2023)). These promising results are at the cost of human-provided demonstrations, which is time expensive. The resulting grasping policies' adaptation capabilities are thus limited by the provided examples.

The acquisition of demonstrations that can bootstrap learning is thus a key matter for solving grasping. Ideally, those demonstrations should be acquired in simulation only (to avoid the issues raised by long-term self-supervised learning on real robots (Levine et al. (2018))), generated with limited human intervention, suited to many grasping scenes (robot, end effector, and objects), and diverse enough to foster generalization capabilities of the learned policies. In this work, **we leverage QD algorithms to generate large sets of diverse grasping trajectories.** The presented results show that a QD method can successfully generate grasping datasets for **different end effectors and objects.** Plus, the optimized fitness functions **associate to each generated grasp a quality label** that can be straightforwardly used for training.

3 Problem

3.1 Notations

This section introduces the notations used throughout this paper. The following subsections describe the QD notations background (section 3.1.1), the notion of fitness sparsity (section 3.1.2), behavioral sparsity (section 3.1.3), and the notations related to policy learning applied to robotics (section 3.1.4). Figure 2 overviews the notations.

3.1.1 Background

This work relies on QD standard notations (Cully et al. (2022)). Let $\Theta \subseteq \mathbb{R}^{n_\theta}$ be the *parameters space*, $\theta \in \Theta$ an *individual* (also referred as a *genome* or a *solution*). Let $\mathcal{B} \subseteq \mathbb{R}^{n_b}$ be the *behavior space*. We note $\phi_{\mathcal{B}} : \Theta \rightarrow \mathcal{B}$ the *behavior function*, such that $b_\theta = \phi_{\mathcal{B}}(\theta)$ is the *behavior descriptor* of θ . Let $f : \Theta \rightarrow \mathbb{R}$ be the *fitness function*, $d_{\mathcal{B}} : \mathcal{B}^2 \rightarrow \mathbb{R}$ a distance function within $\mathcal{B}$. We aim to generate an *archive* A defined as:

$$\left\{ \begin{array}{l} \forall b \in \mathcal{B}_{reach}, \exists \theta \in A, d_{\mathcal{B}}(\phi_{\mathcal{B}}(\theta), b) < \epsilon \\ \forall \theta' \in A, \theta' = \operatorname{argmax}_{\theta \in N(b_{\theta'})} f(\theta) \end{array} \right. \quad (1)$$

where $\mathcal{B}_{reach} \subseteq \mathcal{B}$ is the *space of reachable behaviors*, $\epsilon \in \mathbb{R}^{+*}$ is a small value that defines the density of $\mathcal{B}_{reach}$ paving, and $N(b_{\theta'}) = \{\theta \mid neighbor_{d_{\mathcal{B}}}(b_{\theta'}, b_{\theta'})\}$ is the set of solutions for

As the QD objective is usually to illuminate a behavioral space (i.e. fill $\mathcal{A}$ with high-performing solutions), we can see those problems as looking for a way to explore $\mathcal{B}$ while sampling parameters from Θ , considering that $\phi_{\mathcal{B}}$ is unknown and usually highly non-linear (Doncieux et al. (2019)). The exploration of $\mathcal{B}$ – and the concurrent optimization of f – is carried through the evaluations of a *domain* (or *environment*). This step is described in subsection 3.1.4 through the prism of policy learning in robotics.

This work focuses on the sparse fitness context. In grasping, most of the evaluated θ_i result in $f(\theta_i) = 0$. Let $f_c : \Theta \rightarrow \{0, 1\}$ be the sparse *success criterion*, such that $f_c(\theta) = \mathbb{1}_{f(\theta) > 0}$. In this context, the actual output of QD is a *success archive* A_s defined as $A_s = \{\theta \in A_o \mid f_c(\theta) = 1\}$. Thus, the space to explore is the *goal space* $\mathcal{G}$ defined as $\mathcal{G} = \{b_\theta \in \mathcal{B} \mid f(\theta) > 0\}$, and more accurately the *space of reachable goals* $\mathcal{G}_{reach} = \{b_\theta \in \mathcal{B}_{reach} \mid f(\theta) > 0\}$.

This paper argues that applying QD to grasping requires facing a sparse interaction problem: $\exists \theta \in \Theta$ such that $\phi_{\mathcal{B}}(\theta)$ is not defined. We say here that θ results in a *non-eligible*

behavior descriptor². The set of *eligible* behavior descriptors $\mathcal{B}_{\text{elig}}$ are defined as $\mathcal{B}_{\text{elig}} = \{b \mid \exists \theta \in \Theta, b = \phi_{\mathcal{B}}(\theta)\}$. In this study, $\mathcal{B}_{\text{elig}} = \mathcal{B}_{\text{reach}}$ (see section 3.3.2).

3.1.4 Policy learning in Robotics

We evaluate a policy π_{θ} on the Markov Decision Process $\mathcal{M}$, corresponding to the task's environment. We then obtain a trajectory $\tau_{\theta} \in S_{\tau}$, defined as a sequence of states and actions for each evaluation step along the *episode*. S_{τ} is usually in high dimension. We then project τ_{θ} from S_{τ} to a space $\mathcal{B}$ s.t. $\dim(\mathcal{B}) \ll \dim(S_{\tau})$, and uses N to compare resulting behaviors. We here consider a fixed initial state s_0 and a deterministic $\mathcal{M}$.

In practice, the behavior function $\phi_{\mathcal{B}}$ is the result of the interaction of π_{θ} with $\mathcal{M}$, and the projection in $\mathcal{B}$ of the resulting trajectory τ_{θ} . To better match the conditions of present work's experiments, we will consider that the evaluation of a policy π_{θ} lead to a point τ_{θ} in the trajectory space S_{τ} , and the above-defined fitness functions project elements of S_{τ} to their respective space: $\phi_{\mathcal{B}} : S_{\tau} \rightarrow \mathcal{B}$ and $f : S_{\tau} \rightarrow \mathbb{R}$.

3.2 Taxonomy of QD methods

Quality Diversity (QD) and Novelty Search (NS) are concepts that describe specific methods and imply the emergence of specific addressable problems. From an algorithmic point of view, there is a clear difference between those two families of methods: NS does not rely on a fitness function – while QD does.

Nevertheless, those notions might easily be mixed up. Many works include vanilla NS among tested methods that derivate from NSLC and MAP-Elites (Pugh et al. (2016), Paolo et al. (2021)), while some others introduce as NS-related some methods that match the above-mentioned definition of QD (Kim et al. (2021), Morel et al. (2022)). The commonly shared definition of QD – an algorithm that maintains diversity and optimizes a local quality criterion – is itself not restrictive enough to prevent NS from being considered as a specific instance of QD algorithm³.

We believe that some key insights can result from an exhaustive comparison of QD methods of different nature. Such an approach requires to avoid the mentioned ambiguities. To make the analysis easier, **we propose a taxonomy that allows us to focus on each of the similarities and differences between those methods.** Actually, the ambiguities come from the usage of *NS-based* and *QD-based* methods for different purposes. Here is a list of the main reasons for the ambiguous usage of those notions and the alternative we propose to clarify them:

- **Root algorithm.** The most well-spread misuse of language on that matter is to distinguish algorithms derivated from MAP-Elites (Mouret and Clune (2015)) and those derivated from Novelty Search (Lehman and Stanley (2011a)) by respectively calling them NS-based methods and QD-based methods. The implicit idea behind this shortcut is that those derivated algorithms share common properties of the root algorithm. The problem is that we cannot state if the shared property is the nature of the archive, the way the population is generated, or even the overall goal. In practice, many algorithms share common properties with NS (Lehman and Stanley (2011b), Kim et al. (2021), Morel et al. (2022)) or with MAP-Elites (Bruneton et al. (2019), Nilsson and Cully (2021), Macé et al. (2023)). We will thus refer to those families of methods as *NS-derivated* and *ME-derivated*, using any of the below-proposed distinctions as needed to avoid confusion.

²In practice, many QD algorithms require each individual to have a defined behavior descriptor. To avoid this issue, we set non-eligible descriptors to a vector of 0. For that reason, we can refer to the search space region in which the resulting behavior is eligible as the *support* of $\phi_{\mathcal{B}}$.

³NS novelty can be actually be considered as a dynamic quality criterion the algorithm is optimizing by maintaining a diversity of solutions throughout the evolutionary process.

- **Nature of the archive.** One might talk about NS-based methods for unstructured-archive-based ones, and QD-based methods for structured-archive based ones (Cully et al. (2022)). The structured archive is here expressing the container based on a grid (Mouret and Clune (2015)) from those in which the novelty is computed with a nearest-neighbors approach (Lehman and Stanley (2011a)). Note that the introduction of CVT-MAP-Elites (Vassiliades et al. (2017)) and the notion of minimal novelty to add an individual into an archive (Lehman and Stanley (2010)) makes those containers work similarly. To clarify this matter, we will always use the notions of *UA-based* (Unstructured-Archive) and *SA-based* (structured-archive) methods to distinguish them.
- **Population.** Considering that UA-based methods result from NSLC (Lehman and Stanley (2011b)) and SA-based from MAP-Elites (Mouret and Clune (2015)), one might use the notion of NS-based methods for population-based one – in which a living population is maintained concurrently with a container – and QD-based methods for non-population-based one – in which offspring are generated from individuals directly sampled from the container. But nothing prevents NS-derived methods from relying on the container to sample parents. An explicit distinction should thus be made between *PP-based* methods (population-based) and *NPP-based* methods (non-population-based).
- **Run goal.** Another point of ambiguity is the purpose of the algorithm execution itself. While the ultimate goal of QD methods is clearly established (i.e. generate a set of diverse and high-performing solutions – here referred to as *Rapid Illumination of a Behavior Space (RIBS)*), NS-derived methods aim to uniformly cover a behavior space (referred here as *Behavior Space Coverage, BSC*) (Wiegand (2020), Doncieux et al. (2019)) and by doing so, find an optimal solution to a given problem (*single objective optimization, SOO*) (Lehman and Stanley (2011a), Shorten and Nitschke (2014)).

Finally, it is worth noting that the notion of Quality-Diversity becomes the most well-spread terminology when referring to the overall field. We will thus use *QD* and *QD-based* methods to talk about the field in general and distinguish it from other research perspectives on policy learning.

This work considers state-of-the-art QD methods to generate a diverse set of high-performing grasping trajectories. To do so, **we compare methods with any of the abovementioned mechanisms**, considering that **some could have been designed to address another primary purpose but might lead to promising results or properties on the considered task.**

3.3 Behavioral characterization

Grasping is a challenging task (see section 2.3) in which the definition of the behavior space $\mathcal{B}$ is not trivial. This section motivates the behavioral characterization used in this work experiment. At first are introduced the concepts and notions that can help to define a behavior space $\mathcal{B}$ for a new problem to address (section 3.3.1). These concepts are then applied to grasping, in which some facilitating behavioral hypotheses cannot be verified (section 3.3.2).

3.3.1 Defining $\mathcal{B}$ for a QD problem

This subsection provides definitions of key behavioral concepts. These notions allow to define a suited $\mathcal{B}$ for a given problem but also to stress the challenges caused by its behavioral characterization. At first are defined the notions of *driving* and *describing* $\mathcal{B}$ (section 3.3.1.1), followed by the notion of *task alignment* (section 3.3.1.2), then the *density* of a behavioral function (section 3.3.1.3) and finally the notion of *behavioral completeness* (section 3.3.1.4).

3.3.1.1 Driving $\mathcal{B}$ vs Describing $\mathcal{B}$.

In section 2.2, we presented the history of NS-QD methods and how the two families of methods resulted in different paradigms. Depending on the targeted overall goal, we can distinguish two main usages of the behavior space. When doing SOO or BSC, the behavior space guides the evolutionary process toward the optimal solutions or the exploration of an outcome space. In those cases, $\mathcal{B}$ plays a *driving* role. Introduced in the seminal Novelty-Search paper (Lehman and Stanley (2011a)), this idea is still critical in recent works on NS-derived methods (Paolo et al. (2021)). When doing RIBS, the behavior space depicts how diverse is a solution θ compared to θ' for the targeted task – through the comparison of b_θ and $b_{\theta'}$, therefore playing a *describing* role. This idea can first be found in NSLC (Lehman and Stanley (2011b)) and MAP-Elites (Mouret and Clune (2015)) papers, and is critical for all recent works on ME-derived methods (Fontaine and Nikolaidis (2023), Anne and Mouret (2023)). In brief, **a driving $\mathcal{B}$ helps to discover solutions, while a describing $\mathcal{B}$ allows to distinguish them**. In any case, $\mathcal{B}$ is always driving and describing, but algorithms almost always focus on one or the other usage of $\mathcal{B}$.

It is worth noting that some recent works proposed to explore multiple behavior spaces (*multiBD*) to leverage the two usages of $\mathcal{B}$: Kim et al. (2021) explores both the last position of the ball thrown by the robot (driving $\mathcal{B}$) and the orientation of the end effector at the middle of the episode to generate diverse ways to throw the ball to a given position (describing $\mathcal{B}$). Similarly, NSMBS (Morel et al. (2022)) explores several behavior spaces to generate diverse grasps (describing $\mathcal{B}$) while also exploring the position of the object at the end of the episode to force the generation of successful grasp (driving $\mathcal{B}$). The multiBD paradigm raises many questions: How to explore several behavioral spaces efficiently? How to design or learn the most relevant driving or describing $\mathcal{B}$? More importantly, how to compare QD methods on the obtained results, and how to interpret the output of the algorithm?

This work studies sparse reward and interaction through the application of grasping in robotics. As the ultimate objective is to do RIBS, **a good describing $\mathcal{B}$ is required**. As grasping is a sparse reward task, **a good driving $\mathcal{B}$ is also required**. To let the abovementioned multiBD questions for future work, **the problem must be addressed through a single $\mathcal{B}$ that is both driving and describing**.

3.3.1.2 Task alignment

In QD methods, the behavior space $\mathcal{B}$ supports the exploration – either to push the solutions toward some part of the outcome space (driving $\mathcal{B}$) or to distinguish them (describing $\mathcal{B}$). Several works discussed the importance of **having a good driving $\mathcal{B}$ for making those methods successful**. In particular, Pugh et al. (2015) shows that the success of QD methods requires that $\mathcal{B}$ must be “aligned with the notion of quality”.

QD methods succeed on hard exploration problems because the exploration of $\mathcal{B}$ guarantees to eventually find a successful solution. This matter has been discussed through the hypothesis of uniform sampling of $\mathcal{B}_{reach}$ by Doncieux et al. (2019). We propose to merge the idea of alignment with the hypothesis of uniform sampling through the following definition:

Definition 1 A behavior space $\mathcal{B}$ is aligned with a task submitted to a fitness function f , a success threshold f_s , and a goal space $\mathcal{G} = \{b_\theta \in \mathcal{B} \mid f(\tau_\theta) > f_s\}$ if the probability p_{θ^s} to find a solution θ^s such that $b_{\theta^s} \in \mathcal{G}$ verifies $\lim_{n_e \rightarrow \infty} p_{\theta^s} = 1$, where n_e is the number of domain evaluations.

Let us illustrate this idea with the experimental example given by Doncieux et al. (2019): the two wheels navigation robot that has to reach a specific point of a given maze at the end of the episode. Let us assume the hypothesis of NS uniform coverage proposed in the paper.

The present work argues that **a complete behavioral characterization cannot be trivially defined for many interesting tasks**. The next section elaborates on **why grasping is one of them**.

3.3.2 Defining $\mathcal{B}$ for grasping:

The definition for $\mathcal{B}$ is crucial for making QD algorithms work well (Pugh et al. (2015)). The ideal $\mathcal{B}$ would be: 1) in **low dimension** (Cully et al. (2022)), 2) **unique** (to avoid the complexity of multiBD), 3) **aligned with the task** (to efficiently drive the exploration), 4) **meaningful from the task perspective** (to generate a diversity of solution we are interested in), 5) and **easily interpretable**. This section elaborates on why misalignment cannot be avoided in grasping, and propose a behavior space that satisfies all the other mentioned criteria.

3.3.2.1 Why misalignment cannot be avoided

To address hard exploration problems, QD works on NS-derived methods that rely on a behavior space $\mathcal{B}$ aligned with the targeted task. In these works, $\mathcal{B}$ is defined such that it can be inferred from the expression of the success criteria f_c . In Paolo et al. (2021), all the success criteria depend on the last position of the object of interest (e.g. the end effector, a ball). $\mathcal{B}$ is thus aligned with the task: the exhaustive exploration of $\mathcal{B}$ defined as the agent's last position will eventually result in the discovery of the best-performing solutions.

A similar approach might be applied for grasping: infer from f the trajectory's components that must be explored in order to find the best solutions. This work considers a grasping trajectory to be successful if a validation condition $\xi : \mathbb{R}^6 \times \mathbb{R}^6 \rightarrow \{0, 1\}$ is verified for N_g steps:

$$\left(\sum_{i=1}^{N_g} \mathbb{1}_{\xi(X_{obj}^{T-N_g+i}, X_a^{T-N_g+i})} \right) = N_g$$

where X_{obj}^i and X_{ee}^i are respectively the state of the object and the agent's end effector at step i . Both states are expressed as a 6 degrees-of-freedom pose, that is, the concatenation of a Cartesian position and its orientation in Euler angles. The grasping validation condition ξ is verified if the end effector and the object are in contacts⁴, and if $z_{obj}^i > z_{obj}^0$, z_{obj}^i being the z component of the object's Cartesian position at step i . Inferring $\mathcal{B}$ from the above success criterion lead to:

$$\dim(\mathcal{B}) = N_g(\dim(X_a) + \dim(X_{obj}))$$

By setting $N_g = 10$ steps, as used in this work experiments:

$$\dim(\mathcal{B}) = 10 \times (6 + 6) = 120$$

The obtained value is way larger than low dimensional behavior spaces usually considered in QD, where $\dim(\mathcal{B}) < 10$. QD methods are designed to operate on low dimensional spaces (Cully et al. (2022)). Considering such a high dimensional $\dim(\mathcal{B})$ raises many questions: Can QD methods correctly explore such a large space? Should the obtained behavioral vector be encoded in a lower dimension – falling into the problems of representation-learning for QD (Cully (2019), Paolo et al. (2020))? Note that this $\mathcal{B}$ is designed to drive the exploration. Should a describing $\mathcal{B}$ also be defined – resulting in a multiBD problem?

⁴Assuming the 3D models of the object and the robot are known, the contacts can be detected with the 6 degrees-of-freedom poses only.

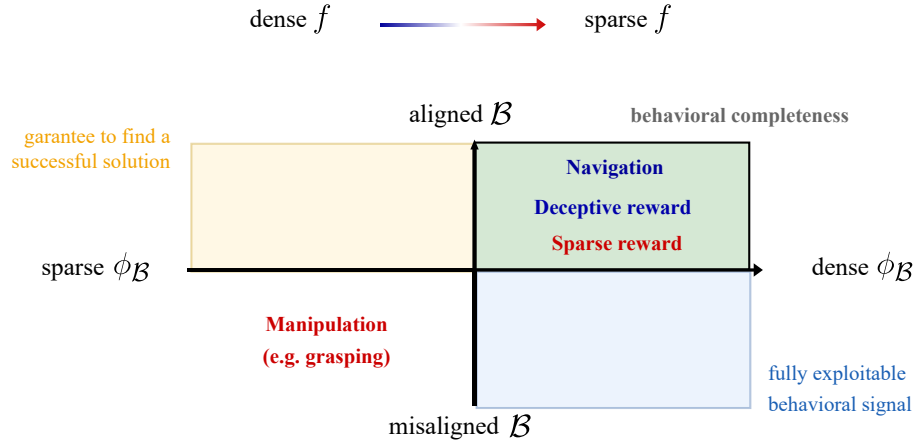


Figure 3: **Challenges to address when using QD for grasping.** In standard QD benchmarks for evolutionary robotics, the behavioral characterization is complete: $\phi_{\mathcal{B}}$ is dense, and $\mathcal{B}$ is aligned with the task. This is true for rapid illumination of $\mathcal{B}$ in locomotion tasks (Cully et al. (2022)), deceptive reward problems (Lehman and Stanley (2011a)), or sparse reward QD (Paolo et al. (2021)). By studying grasping, the present work argues that addressing QD with manipulation tasks can lead to sparse $\phi_{\mathcal{B}}$ and misaligned $\mathcal{B}$.

The grasping controller consists of an open-loop trajectory guided by 3 waypoints. The gripper position is initialized to open position. When the end effector first touches the object, the gripper is closed with constant force. This mechanism is inspired by the *Palmar Grasp Reflex*, which makes newborn infants close their hands when pressure and touch are applied to the palm (Futagi et al. (2012)). **The controller is described in detail in the section A of the supplementary materials.** Finally, the fitness function consists of a normalized mixture of two sub-fitnesses that aim to minimize energy consumption and to minimize the variance of contact points between the end effector and the objects. Note that its value is set to 0 if the object is not grasped. **The fitness function computation is detailed in section B of the supplementary materials.**

4.2 Methods

Table 1 provides an overview of each of the studied methods with respect to the taxonomy proposed in section 3.2. This table shows why NS and ME-derived methods are usually called NS and QD-based methods, as most of those algorithms share common properties. However, we believe this matter is a chicken-and-egg problem: the lack of accurate distinction results in implicit design choices that do not question previously established algorithmic paradigms.

This section provides details on the compared methods. All methods rely on the same behavioral characterization ($\phi_{\mathcal{B}}(\tau) = X_a^{touch}$). When not explicitly stated, the mutation is a Gaussian perturbation. We decided not to use the crossover to avoid adding too many complexities and hyperparameters that might affect the results. Some works in the field suggest that this could improve the performances (Vassiliades and Mouret (2018)); crossover is thus considered for future work. For comparison fairness, all NPP methods that can sample a variable number of individuals from the container are implemented such that this number matches the population size of PP methods.

methods	root			container			pop		goal		
	NS	ME	$\emptyset$	UA	SA	$\emptyset$	PP	NPP	RIBS	cvg($\mathcal{B}$)	θ^*
Random			•			•	•				
NS	•			•			•			•	•
Fit			•	•			•				•
NSLC	•			•			•		•		
NSMBS	•			•			•			•	
SERENE	•			•			•		•		
ME-rand		•			•			•	•		
ME-scs		•			•			•	•		
CMA-ES			•			•	•				•
CMA-ME		•			•			•	•		
CMA-MAE		•						•	•		

Table 1: **Taxonomy of the compared methods.** Each column corresponds to a family of QD method (see section 3.2): the root algorithm (Novelty Search (NS), MAP-Elites (ME), or none of those two ($\emptyset$)), the nature of the container (unstructured archive (UA), structured archive (SA), or no container ($\emptyset$)), the population mechanism (population-based (PP) or directly sampled from the archive (NPP)), and the overall objective of the algorithm (rapid illumination of a behavior space (RIBS), dense coverage of a behavior space (cvg($\mathcal{B}$)), or finding the optimal solution (θ^*)).

(Paolo et al. (2021)). The method can be decomposed into two phases: an exploration phase consisting of a standard NS algorithm; and an exploitation phase, exploiting solutions found with non-null reward to initialize CMA-ES emitters to refine the solutions. To our knowledge, this is the only work specifically focusing on sparse reward context for QD.

CMA-ES. Covariance Matrix Adaptation Evolution Strategy (CMA-ES) is one of the best derivate-free optimization algorithms for continuous domains (Hansen (2016)). It models the sampling distribution of the population as a multivariate normal distribution, estimated from the previous generation’s best-performing solutions. Even though this method is single objective-oriented, its impressive results on many problems made us use it as a baseline to estimate how good other methods are to explore, or to generate high-performing solutions. Note that CMA-ES is not a QD algorithm.

CMA-ME. Covariance Matrix Adaptation MAP-Elites (CMA-ME) combines self-adaptation techniques of CMA-ES with diversity-maintaining techniques of MAP-Elites (Fontaine et al. (2020)). Despite the great results shown in the paper, recent work shows that this method struggle in sparse reward domains (Paolo et al. (2021)). This method is kept as a baseline to verify this weakness in new domains and emphasize key properties of other algorithms.

CMA-MAE. Improved version of CMA-ME to address three of its limitations: quick abandonment of difficult-to-optimize objectives, inability to efficiently explore flat objective functions, and inefficiency on low-resolution archives. The results presented in this recently published paper (Fontaine and Nikolaidis (2023)) makes CMA-MAE a state-of-the-art QD method. The fact that it has been explicitly designed to address flat fitness landscape scenarios makes it a serious candidate to tackle grasping.

Given the good results of MAP-Elites, a study has also been conducted on the following variants:

ME-fit. Standard MAP-Elites that select solutions with respect to their fitness, sorted in descending order.

ME-nov. Standard MAP-Elites that select solutions with respect to their novelty, sorted in

Evolutionary Computation Volume x, Number x

4.3.1 Algorithm output

As most QD works focus on either NS-derived or ME-derived methods, the running output is usually the container used by algorithms. RIBS-related works measure the structured archive coverage and qd-score (Fontaine and Nikolaidis (2023), Macé et al. (2023)), while all works that focus on NS measure performances on the resulting unstructured archive or extracting from the evolutionary process the best-performing individual (Paolo et al. (2021), Conti et al. (2018)).

Comparing QD methods of different natures raises many issues: is the qd-score still a relevant metric when comparing SA and UA-based algorithms? How to compute $cvg(\mathcal{B})$? More generally, can the evaluation of performances be independent from the container used during the evolutionary process – discarding the constraint of operating in a fixed behavior space or even the requirement of using one? Recent works in the QD community stressed the rising need for distinguishing the output of the algorithms from the algorithmic modules themselves (Fontaine and Nikolaidis (2023)).

In this work, **we introduce an evaluation procedure that distinguishes the algorithms' output from its internal components.** The evaluation framework is described in Figure 5, and basically consist of an external MAP-Elites that does not generate any new solution. The interaction between the environment and the QD algorithm is considered as a black box that generates solutions θ_i for a given budget b . All the solutions that have ever been generated during a given run are submitted to this evaluation procedure. Each obtained trajectory τ_{θ_i} is then then projected into a behavior space $\mathcal{B}$, obtaining a behavior descriptor b_{θ_i} . This descriptor is then considered for being added into an *outcome archive* A_o , similarly to a standard MAP-Elites. The *success archive* A_s is then defined as : $A_s = \{b_{\theta_i} \in A_o \mid f(\tau_{\theta_i}) > 0\}$. Note that in the case of non-sparse reward domains, $A_s = A_o$.

4.3.2 Metrics

Considering the above-proposed evaluation framework, the analysis is conducted on the following metrics:

Coverage of the Outcome Archive. As the behavior space is here $\phi_{\mathcal{B}}(\tau) = X_a^{touch}$, computing the coverage of A_o results here to answer the question: how diverse are the entry point for all the grasping attempts? The higher $cvg(A_o)$, the more first touching points have been discovered by the agent.

Coverage of the Success Archive. Computing $cvg(A_s)$ answers the following question: How many diverse successful grasps have been found? Interestingly, this metric appeared to be the most important one, as the qd-score for the chosen fitness function is aligned with this metric (see supplementary materials C).

Top-N fitnesses. To get an idea of how efficient the generated solutions are, the top-N fitnesses ever produced by each algorithm are also compared. This metric is an alternative to the qd-score to estimate the quality of the diverse generated solutions. Qd-score might be dominated by the number of found solutions, especially in difficult exploration tasks like grasping. It informs us on the number of successful grasps found, but not on their performances.

Environment difficulty. To evaluate the challenges associated with each studied task, the environment difficulty is evaluated through two metrics: the *outcome ratio* η_o and the *success ratio* η_s . To compute those metrics, N_{ed} random individuals θ_i are generated by sampling from a uniform distribution within the genotype space Θ . The obtained θ_i are then evaluated on the environment $\mathcal{M}$, getting the resulting number of individuals that successfully touched the object (n_o) and the number that grasped it (n_s). The ratios are then computed as follows:

$$\eta_o = \frac{n_o}{N_{ed}} \quad \eta_s = \frac{n_s}{N_{ed}}$$

Those metrics stress whether an environment is submitted to sparse behavior function

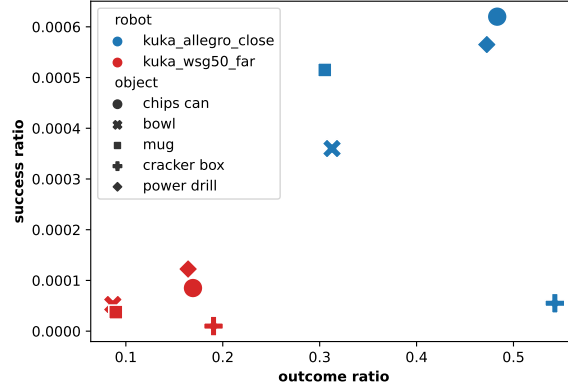


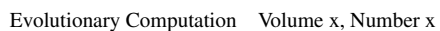
Figure 6: **How challenging are the studied tasks.** We indeed are in a sparse reward context, as a few of the sampled trajectories lead to a non-null fitness. Plus, many deployed trajectories do not provide any behavioral information to exploit, making the exploration even more difficult. The higher the outcome ratio, the higher the success (Pearson correlation $r = 0.61$ with $p < 0.06$). Overall, *kuka_allegro_close* domain is easier than *kuka_wsg50_far* – except for the Allegro on the cracker box, in which the object is easy to touch but hard to grasp.

The section 5.1 presents the result on the difficulties associated with each domain. The results obtained on 100k evaluations are provided in section 5.2. The results obtained on 400k evaluations are provided in section 5.3.

5.1 Environment difficulty

Figure 6 gives an overview of the challenges associated with each domain. The outcome ratio on *kuka_allegro_close* is significantly higher than those obtained on the *kuka_wsg50_far*: between 31% and 54% of the randomly sampled trajectories touches the object on the first domain, while between 9% and 19% on the other one. In both cases, **many trajectories do not provide a behavioral signal that the algorithm can exploit to guide the exploration toward high-performing solutions**. As expected, the smaller objects (bowl and mug) lead to lower η_o than the larger ones (power drill, cracker box, and chips can). **All domains are submitted to sparse rewards**, as the higher value of η_s is close to 0.06% of success, meaning that in a pure random search context, about 99.94% of trajectories result in a null fitness.

Overall, the domains in which the object is easier to touch are those in which the object is easier to grasp (Pearson correlation between η_o and η_s : $r = 0.61$ with $p < 0.06$). The Allegro hand on the cracker box is a remarkable outlier, though: it is easier to touch it ($\eta_o = 0.54$) than any other objects on that gripper, but it is about 10 times more difficult to grasp it than on any other tested objects. The easier domains are (*kuka_allegro_close*, chips can) and (*kuka_allegro_close*, power drill), with respectively 48% and 47% of chance to randomly touch the object, and about 0.062% and 0.057% of chance to grasp it. (*kuka_allegro_close*, bowl) can be seen as a intermediate-difficulty environment ($\eta_o = 0.31$ and $\eta_s = 0.00036$). The most challenging domains are the *kuka_wsg50_far*-based setups, especially on the cracker box on which the success ratio reaches its smallest value ($\eta_s = 0.00001$). The most challenging environment regarding the behavioral sparsity are (*kuka_wsg50_far*, mug) and (*kuka_wsg50_far*, bowl). It is also worth noting that a similar difficulty pattern can be seen on both robots for the



studied object: the chips can and the power drill are the easier objects to address; the cracker box and the mug are more challenging tasks regarding both behavioral and fitness sparsity; and the cracker box is the most difficult to grasp object – despite of its high probability to touch it. The ratios obtained on the cracker box can be explained by the fact that there is a high probability of making the object fall, resulting in an impossible-to-grasp state for those grippers – while the chips can and the power drill are less challenging as they can still be grasped after having made them fell on the table.

The next section presents the result obtained after 100k evaluations on the compared methods.

Generation of diverse successful solutions. In Figure 7 are the obtained coverage of the success archive for each method throughout the evolutionary process. The most striking result is that **ME-scs is dominating all the compared methods on all the evaluated domains by a large margin**. CMA-MAE outperforms all other methods except for ME-rand, which reaches similar coverage for several environments. SERENE is way below, and so is NSMBS. Another interesting point is that NSLC, the NS-derived RIBS method, does not succeed as well as the ME-derived variant. We attribute this to its selection process and its PP nature. On such a hard exploration problem, it is difficult to reach non-null fitness – such that the local quality score (Lehman and Stanley (2011b)) is likely to either be at the maximal value (i.e. the individual is

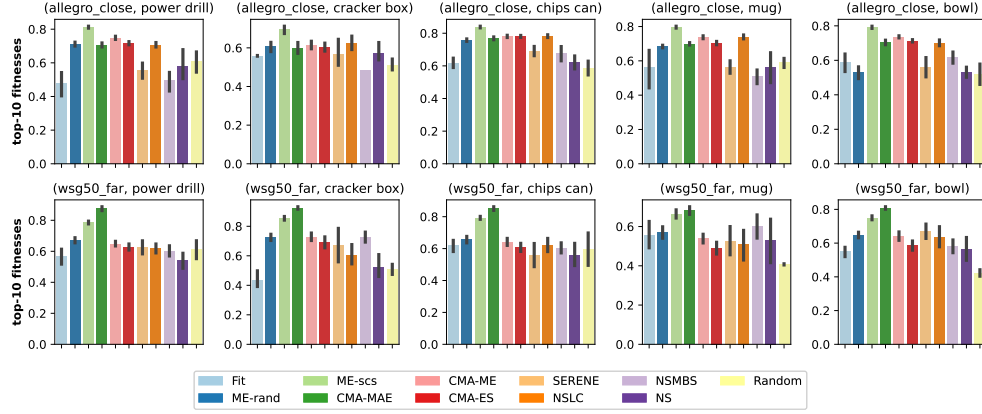


Figure 8: **Fitnesses of the top-10 best performing individuals obtained after 100k evaluations.** ME-scs and CMA-MAE outperforms other methods after this number of evaluation, obtaining high-performing solutions with comparable fitnesses.

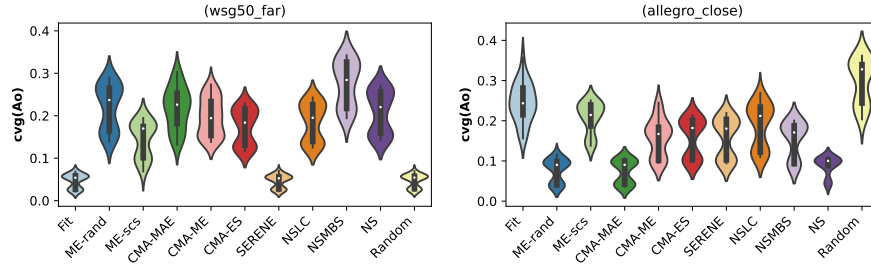


Figure 9: **How do QD methods explore the objects surface.** Coverage of the output archive after 100k evaluations, over 10 seeds. Even though ME-scs crushes other methods on the generation of successful grasps, its exploration of $\mathcal{B}$ is not significantly higher than other methods – meaning that ME-scs prioritize parents that are the more likely to mutate into successful individuals.

card NSMBS from the most promising QD methods to address grasping, stressing what should be its main interest: the multiBD context (see discussion in section 6.1.4).

SERENE reached slightly better performance than NS and got similar results to NSMBS. It matches what has already been discussed in the original paper: SERENE struggles to refine solutions in hard exploration domains (Paolo et al. (2021)). It initially behaves just like a standard NS. As soon as non-null fitness solutions have been found, they are evaluated as candidates for budget assignment to do local optimization. The problem is that in such a hard exploration domain as grasping, a successful solution might not lead to any successful one in a few 1-mutation trials. We attribute the obtained results to this phenomenon: in most cases where NS found successes, the candidates are evaluated as non-promising solutions for local optimization, making the method fall back to a standard NS. It sometimes allows local optimization, which explains the slight performance gain compared to NS on all the domains.

Exploration of the object surface. Figure 9 shows the coverage of the output archive after 100k evaluations. Interestingly, ME-scs does not dominate other methods on that metric. It shows that **by selecting successful solutions from the archive, ME-scs focuses on the regions of the object surface which are more likely to result in successful grasps.**

Surprisingly, methods derivated from NS do not explore the object surface better than other methods. On *kuka_wsg50_far*, NS does not report a higher coverage than most quality-guided methods. Even more surprising is that **NS reports one of the lowest $cvg(A_o)$ on *kuka_allegro_close*.** Pure NS or NS variants that are not often submitted to a non-null quality signal are however optimizing $cvg(\mathcal{B})$ directly. NS-derivated methods which are also guided by quality have generated a significant number of successful solutions: SERENE, NSLC, and NSMBS dominate NS on $cvg(A_s)$ and on the top-10 fitnesses on many domains. They have therefore been confronted to non-null fitnesses, limiting their exploratory capabilities. But no straightforward answer can be brought to NS performances. Results given on the outcome ratio provide hints to explain those results.

Outcome and success ratios. In Figure 10 are displayed the outcome and success ratios measures for each algorithm throughout the evolutionary process on the *kuka_allegro_close* domains. The selection operator of ME-scs leads to a probability of generating success $\eta_s = 3.7\%$. ME-scs is above 60 times more sample efficient than random sampling in Θ on the easier domain and 50 times more sample efficient than the second higher success ratio on that domain (ME-rand). But there is no statistically significant difference between NS and Random on the outcome ratio. Figure 9 shows that NS does not behave as a Random search on this problem, the reported $cvg(A_o)$ is different for both methods. It means that **NS pressure for novelty has a deteriorating impact on $cvg(A_o)$ in this context** – leading to an exploration of $\mathcal{B}$ that is similar to quality-guided methods (*kuka_wsg50_far*) or significantly worse (*kuka_allegro_close*). Such a result goes to the opposite of the strong exploratory power of NS that has been demonstrated in literature so far (Paolo et al. (2021), Doncieux et al. (2019)). **We attribute the deteriorating impact of NS to its pressure for evolvability.** This point is discussed in section 6.2.

The resulting QD-scores are provided in supplementary materials (Figure 15). In this case, this metric is dominated by the number of found successful solutions, preventing us from relying on this measure to evaluate the ability of the tested methods to generate a set of diverse and high-performing solutions. Thus, the results only reflect the size of A_s , which is redundant with the coverage information.

The leading performances obtained with ME-scs invite us to investigate the role of the selection operator on the illumination of A_s through ME variants. This second experiment compares several ME variants on longer runs (400k evaluations). Are also included the second-best performing solution (CMA-MAE) and the anterior version of this method (CMA-ME) to get more clues on how those state-of-the-art approaches behave in those domains.

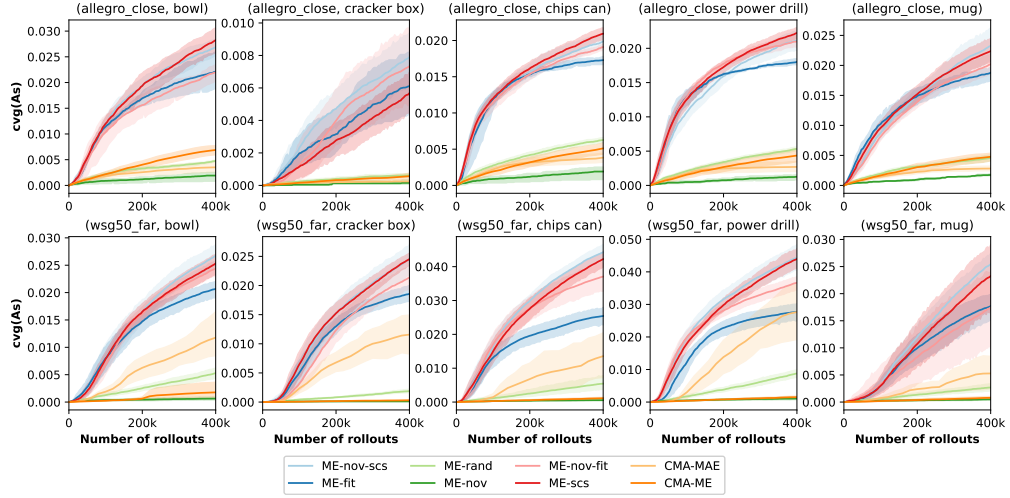


Figure 11: **Impact of the selection operator on ME's ability to generate diverse, successful solutions.** Over 10 seeds. All ME variants that select non-null fitness solution in priority outperforms other methods. Prioritizing fitness can stick the exploration into local minima while selecting successful solutions regardless of their performance increases the success archive's size continuously.

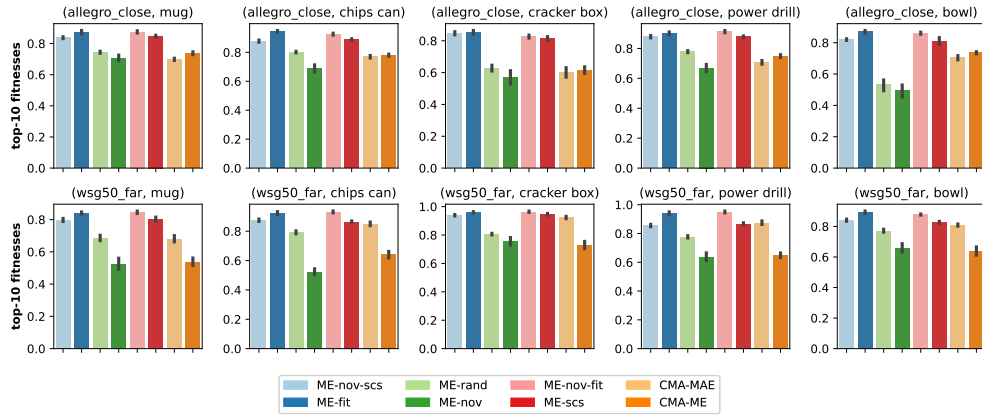


Figure 12: **Impact of the selection operator on high performing solutions fitnesses.** Over 10 seeds, after 400k evaluations. ME-scs outperforms CMA-MAE on longer runs. ME-fit-based variants generate slightly better solutions than ME-scs-based ones.

5.3 Impact of the selection operator

Generation of diverse successful solutions. Figure 11 shows the coverage of the success archive. **Variants that select in priority the successful solutions (ME-scs, ME-nov-scs) dominate all other methods.** Prioritizing the most novel solution among the successful ones does not lead to a statistically significant difference. Selecting individuals with respect to their fitness (ME-fit and ME-nov-fit) leads to better results than other variants, except for the success-based ones. By selecting the best-performing individuals among the already discovered ones, the evolutionary process can get stuck in local minima – preventing the generation of new successful solutions.

Generation of high-performing solutions. Figure 12 shows the obtained distribution of top-10 fitnesses. It can be seen that **ME variants that prioritize fitness lead to the generation of better-performing individuals than success-based selection.** The two ME-scs variants reached comparable results. There are no statistically significant differences between ME-fit variants too. This result is unsurprising: the pressure for novelty is applied to already successful solutions for those 4 variants⁶.

The **poor performances of ME-nov on both $avg(A_s)$ and top-10 fitnesses enforce the result obtained with NS:** pressure for novelty might lead to deteriorated exploration performances. Interestingly, a ME-derived method shares similar properties with an NS-derived one. There are differences between ME-nov and NS beyond the nature of the archive: individuals compete on fitness in their behavioral niche, while NS does not. Discussion on the impact of novelty-based selection under in flat behavior landscape is provided in section 6.2.

CMA-MAE is outperformed by all success or fitness-guided ME variants. It is worth noting that CMA-MAE consistently generates better high-performing solutions than CMA-ME on the most challenging robot (*kuka_wsg50_far*). However, the opposite can be seen on the other robot. The method that dominates this other for generating high-performing solutions is also the one that reached the higher coverage of the success archive (see Figure 11, or more explicitly in section D of supplementary materials). We attribute this observation to the behavior sparsity of the task. CMA-ME is well known for pushing the search away from previously found solutions (Fontaine and Nikolaidis (2023)). It limits CMA-ME from optimizing the performances of already-found solutions. CMA-MAE alleviates this issue with a rolling fitness threshold that conditions elites' insertion into the container. This constraint CMA-MAE's exploration capability, as it spends more budget in the same region of the search space. This mechanism allows CMA-MAE to better exploit previously found successes to find new ones on the most difficult task. On the allegro task, such a strategy might keep the evolutionary process to local minima. CMA-ME exploration capabilities help to escape those minima, while it can be detrimental in a more sparse behavioral domain.

The resulting QD-scores are provided in supplementary materials (section C). Similarly to the first set of experiments (section 5.2), this metric is dominated by the coverage of A_s .

6 Discussion

This section discusses the obtained results – with a spotlight on methods that were expected to get good performances on grasping. Section 6.1 analyses the obtained results on RIBS for grasping. Section 6.2 provides a hypothesis on the detrimental role of novelty in sparse interaction domains, and section 6.3 discusses the proposed evaluation framework.

⁶A multi-objective solution that balances the pressure on performing/successful individuals with the pressure for novelty on non-successful individuals could have been set. As the scope of this work is rapid illumination of a success archive for grasping, the results obtained with novelty-guided solutions are not prone to stimulate too much effort in that research direction.

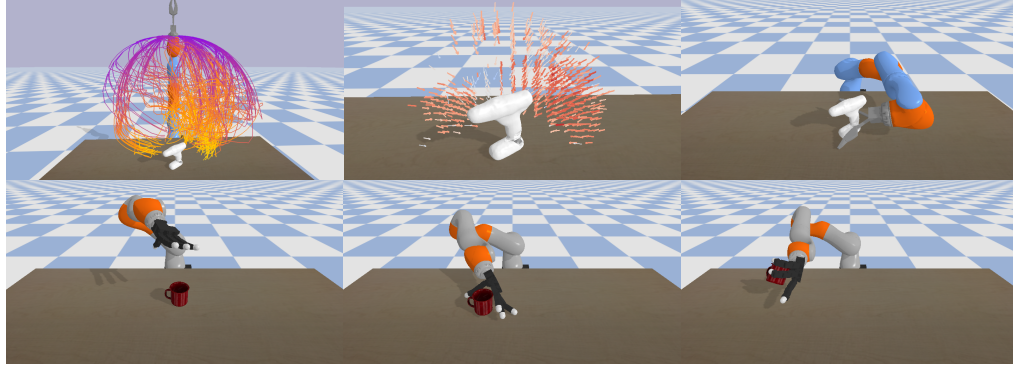


Figure 13: **Output visualizations.** (Top left) 200 trajectories randomly sampled from an A_s produced by ME-scs. Trajectories are displayed as a sequence of end effector positions. Color expresses temporality, from purple to yellow. The generated grasps are spread all over the operational space. (Top center) Visualization of the outcome descriptor associated with each solution from the same A_s . All the 724 solutions are displayed. The hotter the color, the higher the fitness. The regular space between the arrow’s initial points expresses A_s predefined sampling. The best-performing solutions are the ones that minimize the energy cost while maximizing grasp stability. (Top right) Grasping position corresponding to the best-performing individual from the same A_s ($f = 0.95$). (Bottom) A sequence of a randomly sampled success from a generated A_s on the Allegro hand, from left to right.

6.1 Top performing methods vs promising methods

6.1.1 ME-scs

The **high performances of ME-scs variants on grasping is the most striking result** obtained in the above-presented experiments. Grasping rewards greedy exploitation of previously found successes; focusing on local search from successful solutions significantly increases the probability of finding new ones. We can distinguish success-greedy (ME-scs-based) from fitness-greedy (ME-fit-based) variants, respectively focusing on finding new successes or optimizing the quality of already found ones. The gap in performances obtained with those methods compared to QD state-of-the-art shows that **there is room for developing QD algorithms that are suited to address tasks with similar properties.**

6.1.2 SERENE

One can easily create a domain that would show the limits of a ME-scs-like approach, by making the exploration-exploitation tradeoff proposed by SERENE a requirement for success. Such a domain would have the following characteristics: 1) the goal space $\mathcal{G}$ must consist of several distinct regions of $\mathcal{B}_{reach}$ that would require several mutations to move from one sub-region of $\mathcal{G}$ to another one, and 2) making each sub-region of $\mathcal{G}$ large enough to saturate ME-scs’s population with solutions from a single region after having discovered it. Paolo et al.’s redundant arms or curling might be a good example of domains that satisfy those criteria. In this case, SERENE should outperform ME-scs, as its exploration-exploitation mechanisms would eventually lead to the discovery and exploration of each sub-regions of the goal space – while ME-scs would be trapped into the first found sub-region. It would confirm the difference in properties of those two algorithms: **SERENE balances exploration with exploitation**, which might block

its execution into a standard NS algorithm if the rewarding regions are very hard to reach (as discussed in Paolo et al.), while **ME-scs explores until a reward zone is found, and then focuses on the exploitation of this first entry point** to generate as much diversity as possible while concurrently optimizing the reward of each rewarded solutions.

Beyond those theoretical questions, one might ask **what kind of non-toy problems can be addressed by each of those approaches**, considering the above-discussed properties. The present study considers the task of grasping, a key task for both concrete industrial cases (Mészáros et al. (2022)) and open-ended scenarios in robotics (Brohan et al. (2022)). The experimental results suggest that ME-scs’s focus on exploitation is the most promising approach to generating a large repertoire of diverse high-performing grasping solutions. **Similar properties should arise in many other robotics manipulation tasks**, as most of them share some of the challenges studied in the present work. It includes behavioral sparsity, misalignment between the behavioral space and the targeted task, and hard-to-explore localized regions that concentrate the fitness function support.

6.1.3 CMA-MAE

CMA-MAE (Fontaine and Nikolaidis (2023)) is an extension of CMA-ME (Fontaine et al. (2020)) that alleviates some of its weaknesses – its inability to efficiently illuminate a behavior space when facing flat fitness landscapes. This CMA-ME limitation is visible here. While CMA-MAE outperforms most of the compared methods, **its performances are still way below ME-scs variants**. One might argue that **grasping rewards the greedy exploitation of previously found solutions**, as successful individuals lie in a limited part of the genotype space. CMA-MAE would therefore conduct a better exploration-exploitation compromise than a ME-scs method, resulting in lower grasping performances but better generalization capabilities on other tasks. This point can be nuanced by the fact that **grasping is itself submitted to discontinuities that require exploration to mutate one grasp to another successful trajectory that applies forces elsewhere on the object**. It might explain the difference in top-fitness performances for CMA-ME and CMA-MAE on the two tested robots. However, **we cannot state to what extent CMA-MAE struggles in tasks submitted to sparse behavior from those grasping experiments**. As the α parameter of CMA-MAE allows to control its tendency to explore or to behave similarly to a single-objective optimizer, it would be interesting to study how to make CMA-MAE more robust to sparse interaction problems. In particular, how to design QD algorithms that efficiently balance exploration and exploitation in those tasks, as looking for exploration might result in poor exploration.

6.1.4 NSMBS

NSMBS (Morel et al. (2022)) authors introduce their method as a way to address the challenges inherent to the task of grasping. The present work shows that NSMBS is not the only method that can efficiently address grasping. **Our results show that other methods – like ME-scs variants – perform way better on the task**. Indeed, NSMBS’s ability to explore several behavior spaces during the evolutionary process has not been used. One might argue that: 1) making NSMBS evolve on multiple behavior spaces might lead to better results, and 2) that NSMBS’s capacity to optimize several $\mathcal{B}_i$ allows producing an outcome repertoire with a high diversity on multiple components of the trajectory (e.g. how the end effector approaches the object, and how forces are applied on it). The visualization of individuals from a success archive produced by ME-scs (Figure 13) shows that **choosing a single behavior space ($\phi_{\mathcal{B}}(\tau) = X_a^{touch}$) does not result in limited diversity of grasping trajectory**. It is worth noting that the trajectories cover the whole operational space. Exploring the space of object-gripper contact points results in diverse ways to approach the object, as some opposite points on the object’s surface are not likely to be reached with a similar approach – especially if we also optimize the energy consumption through

thanks to the proposed evaluation framework, which is discussed in the next section.

6.3 On the evaluation framework

This experiment shows that the proposed evaluation framework has many advantages for studying QD methods: it is **easy to implement, interpret and visualize**. By distinguishing the evaluation from the algorithmic internal components, **methods based on different kind of archives can fairly be compared**. Such an evaluation framework can similarly be used to compare methods with several behavioral spaces (Kim et al. (2021), Morel et al. (2022)) with QD state-of-the-art approaches. It also has the benefit of **making the outcome of QD algorithms explicit** on a given problem. We hope this approach will be exploited in future QD works to allow more flexibility in algorithm design without compromising the experimental studies' interpretability or exhaustiveness.

A limitation of this evaluation is that it might favor ME-derived variants – as the sampling within the container matches the sampling of the outcome archive. An output structured archive has however many benefits: most of the QD works focus on SA-based methods, and it allows easy visualization, analysis, and exploitation. Practitioners who want to have fine control over granularity according to expert knowledge can set a specific sampling (like in this work), or can rely on a CVT sampling to fix the maximal size of the outcome archive without having to design the grid cell sampling.

7 Conclusion

This work investigates Quality-Diversity (QD) methods under sparse reward and sparse interactions problems through the application of grasping in robotics. What was arguably the most promising QD state-of-the-art methods for this domain (CMA-MAE, SERENE, and NSMBS) are significantly outperformed by variants of MAP-Elites that select non-null fitness solutions in priority. As a result, **an algorithm that is as simple as a standard MAP-Elites appeared to successfully generate a large set of diverse and high-performing solutions on multiple grasping domains**.

The best-performing methods can thus be used to **automatically generate datasets of grasping trajectories**. As the access to demonstrations is a **key matter to solve grasping**, we believe that such dataset generators can provide significant help in the resolution of this task.

Our results suggest that addressing a task submitted to behavioral sparsity can lead to counterintuitive results. In particular, **explicitly guiding an algorithm to optimize the exploration of an outcome space favors selecting the most evolvable and unstable solutions, pushing the offspring out of the behavioral function support**. One might parallel seminal works in Novelty Search and deceptive reward, in which directly optimizing fitness might lead to poor optimization. In this context, **behavior sparsity results in deceptive novelty**.

This paper opens many perspectives for the QD field regarding theoretical and practical matters: Do this work's observations hold in other robotic manipulation tasks? How to design more advanced algorithms that outperform ME-scs variants on grasping? How to avoid the sparse interaction setup through another behavioral characterization without compromising the algorithm's efficiency, as well as the output's interpretability and exploitability? More generally, we hope this work will incite the QD community on evolutionary robotics to tackle more complex problems, which are likely to result in breakthroughs in the field.

8 Acknowledgement

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practice, f_c returns 1 if the following conditions are verified for N_g steps consecutively: 1) this object does not touch the table; 2) the object does not touch the floor; 3) the robot does not touch the table; 4) the end effector of the robot is touching the object; and 5) there is no penetration between the 3D models of the robot and the object. The energy consumption fitness is the sum of the applied torques on each joint throughout the episode. The grasp stability fitness is measured as follows:

$$f_{gs}(\tau_i) = (-1) * \left(\frac{\text{var}(\{X_a^{\text{touch}_{i=1, \dots, T}}\}) + \text{var}(\{X_{obj}^{\text{touch}_{i=1, \dots, T}}\})}{n^{\text{max_touch}} \times n^{\text{cont_touch}}} + \zeta_d \right)$$

with $X_a^{\text{touch}_i}$ being the end effector-object contact point on the agent, $X_{obj}^{\text{touch}_i}$ being the end effector-object contact point on the object, $n^{\text{max_touch}}$ being the number of iterations in which there has been contact between the agent and the object, $n^{\text{cont_touch}}$ being the maximal number of iterations of continuous contact from the first touch between the agent and the object, and ζ_d being an additional cost that penalizes discontinuous interactions. The cost ζ_d is defined as the number of iterations from the first step at which the agent stops to touch the object until the end of the episode. The whole fitness is set as a negative function to maximize. The discontinuity cost is the dominating member of f_{gs} until continuous grasps are found. As f_{gs} cannot be lower than $1 - T$ (meaning that the object is touched during the first step only) and larger than 0, we rely on the assumption that $f_{gs} \in [-T, 0]$ to project it into the expected $[0, 0.5]$ interval. The normalization of f_{ec} relies on extrema values measured from executions of fitness-free methods (Random and NS).

C Qd-scores

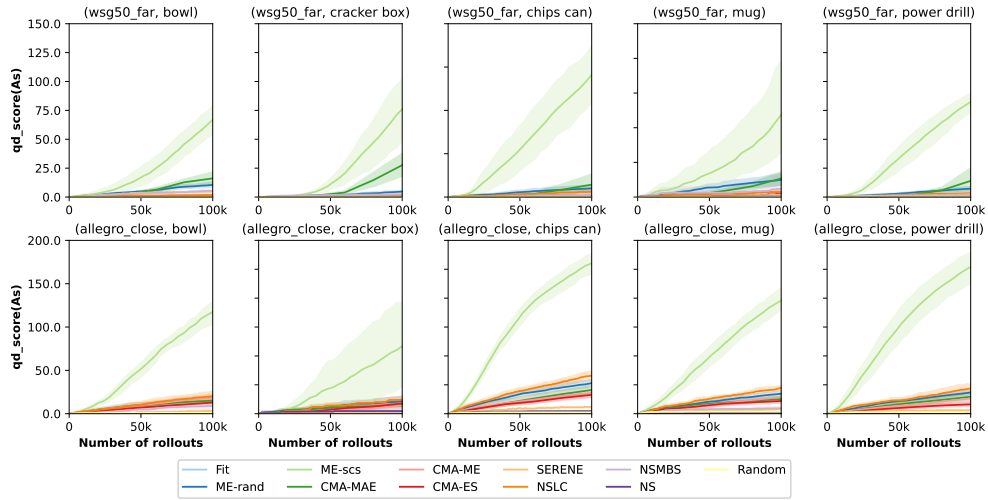


Figure 15: **Qd-scores measured throughout the evolutionary process for state-of-the-art QD algorithms.** The large differences in success archive size make the qd-score saturated by the number of solutions. This measure does not provide information on the quality of the generated successes.

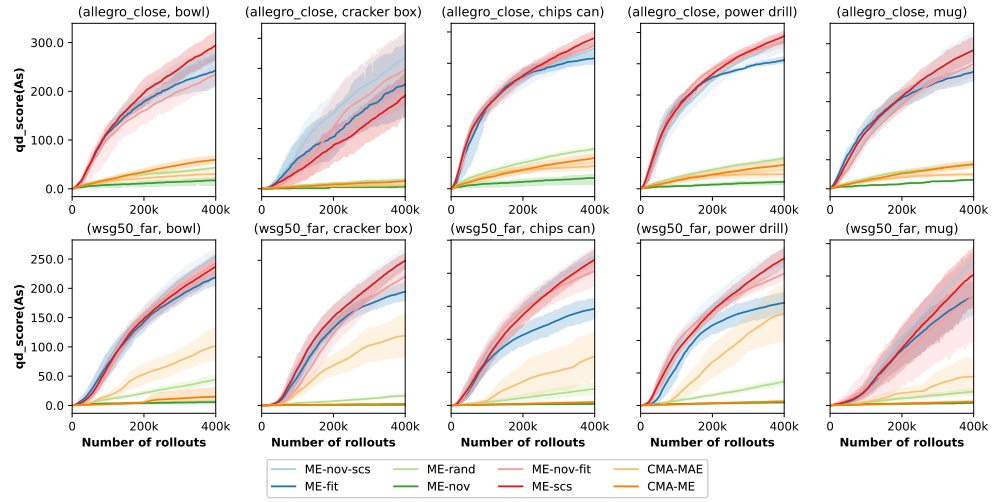


Figure 16: **Qd-scores measured throughout the evolutionary process for different variants of MAP-Elites.** Results are essentially the same as those delivered by the coverage of A_s : fitness-guided ME-* variants can get stuck into local minima.

D Success archive coverage of CMA-* variants

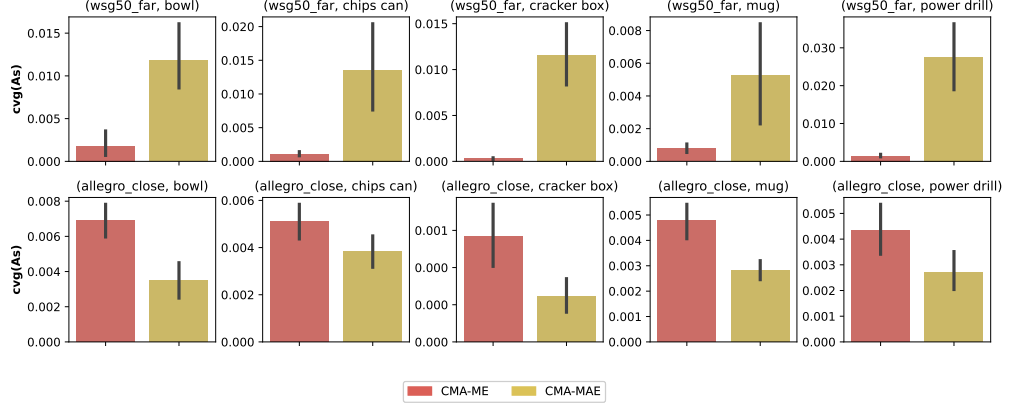


Figure 17: **CMA-ME vs CMA-MAE on the coverage of the success archive** Over 10 seeds, after 400k evaluations. While CMA-MAE explores $\mathcal{G}$ more efficiently on the most difficult task (*kuka_wsg50_far*), CMA-ME is better on the easier one (*kuka_allegro_close*). We attribute this observation to the behavior sparsity of the task. CMA-ME is well known for pushing the search away from previously found solutions (Fontaine and Nikolaidis (2023)). It limits CMA-ME from optimizing the performances of already-found solutions. CMA-MAE alleviates this issue with a rolling fitness threshold that conditions elites' insertion into the container. CMA-MAE spends more budget in the same region of the search space, limiting its exploration capabilities. This mechanism allows CMA-MAE to better exploit previously found successes to find new ones on the most difficult task. Such a strategy might keep the evolutionary process into local minima on the Allegro task. CMA-ME exploration capabilities help to escape those minima, while it can be detrimental under more sparse behavioral domains.